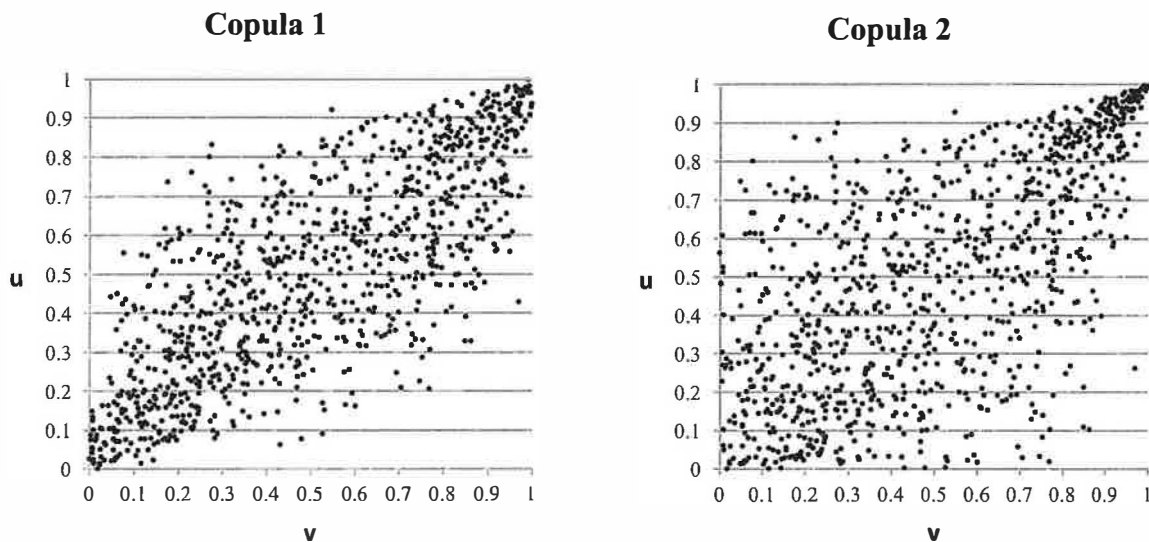


20. (2.25 points)

The following scatter plots were obtained from simulations of two different copulas with the same Kendall τ .



a. (0.75 point)

Identify which copula above is the Normal copula and which is the HRT copula. Fully justify your answer.

b. (0.5 point)

Briefly describe one advantage for each of the two copulas above when used to aggregate the insured losses from multiple lines of business subject to catastrophe events.

c. (1 point)

Explain which of the two copulas above will produce the higher result on the joint distribution for the following two metrics:

- i. Expected value
- ii. $V@R_{0.9}$

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 20	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE(S): C5, C6
SAMPLE ANSWERS	
Part a: 0.75 point	
<u>Sample 1</u> Copula 1 is normal copula as it's more symmetric and has more points bunched up together in the lowest probabilities (0, 0.1), as well as highest probabilities (0.9, 1), indicating tail dependency on both tails, which normal has. Copula 2 is HRT copula as has points bunched up primarily in right tail (0.9, 1.0), implying dependency only in right-tail, which is heavy. <u>Sample 2</u> Copula 1 is the normal copula as it is more uniform across the graph. Copula 2 has higher density in the top corner which indicates an HRT copula. <u>Sample 3</u> Copula 1 is the Normal copula. It is symmetrical, and appears to have lower correlation in the right tail. Copula 2 is HRT. Much higher correlation in the right tail (top right corner).	
Part b: 0.5 point	
<u>Sample 1</u> • HRT copula is advantageous for LOBs that are generally not correlated (workers comp & homeowners) but show high correlation in the cases of extreme events (earthquake). • Normal copula is easier to generalize when we want to join more than 2 lines of business. <u>Sample 2</u> • Normal can handle more than two lines, HRT can't. • HRT has much heavier density in right tail and it is the case that dependency is highest in tail for cat risk. <u>Sample 3</u> • Normal: easy to invert and simulate losses • HRT: CATs are tail events so higher tail weight on right is appropriate	
Part c: 1 point	
<u>Sample 1</u> • Expected value: Same result since copula only influence correlation and thus the variance part. Will not influence expect value. • $VaR_{0.9}$: HRT copula since the risks correlate more under HRT the $VaR_{0.9}$ could be much larger since it captures the loss at tail. <u>Sample 2</u> • Expected value – These should produce the same expected value given that the overall Kendall t value is the same. • $VaR_{0.9}$ – HRT will provide a higher value than Normal, given there is more correlation in the tail for HRT than Normal.	

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Sample 3

- Copula should not change the expected value of the LOBs it is correlating. Correlation impacts variability. Thus, this should be the same.
- HRT -> $\text{VaR}_{0.9}$ is a right tail event. More substantial correlation can be seen at this point in the scatter plot. Higher likelihood at both LOB's being bad thus, HRT.

EXAMINER'S REPORT

Given two scatter plots, candidates were expected to correctly identify which plot is the Normal copula and which is the HRT, to name an advantage of each copula in the case that two lines of business are subject to catastrophe events, to identify which copula will produce the highest joint expected value and joint $\text{VaR}_{0.9}$, and to explain why.

Part a

Candidates were expected to correctly identify which scatter plot corresponds to the Normal copula and which one is the HRT copula.

To obtain full credit, candidates should have correctly identified the two copulas and justified their choice by stating a reason for each copula.

The reasons can be about correlation and/or observations on the scatter plots.

Common mistakes included:

- Misidentification of the copulas
- Same reason given for both copula
- Reason(s) given for HRT copula only

Part b

Candidates were expected to name an advantage of each copula in the case that two lines of business are subject to catastrophe events.

If candidates gave two advantages for one copula, only the first one was graded.

Common mistakes included (Normal copula):

- Talked about the left tail correlation (good for non-cat years)
- Also has a right tail correlation (not an advantage over the HRT for catastrophe events)
- Easy to invert (with no other reason)
- Easy to explain (with no other reason)

Common mistakes included (HRT copula):

- No recurring mistakes were noted by the graders

Part c

Candidates were expected to identify which copula will produce the highest expected value and which one will produce the highest $\text{VaR}_{0.9}$.

Candidates generally performed better on the $\text{VaR}_{0.9}$ part of the question than the expected value part.

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Common mistakes included (expected value):

- Candidates named one copula or the other (the correct response was that choice of copula does not affect expected values)

Common mistakes included ($\text{Var}_{0.9}$):

- No recurring mistakes were noted by the graders